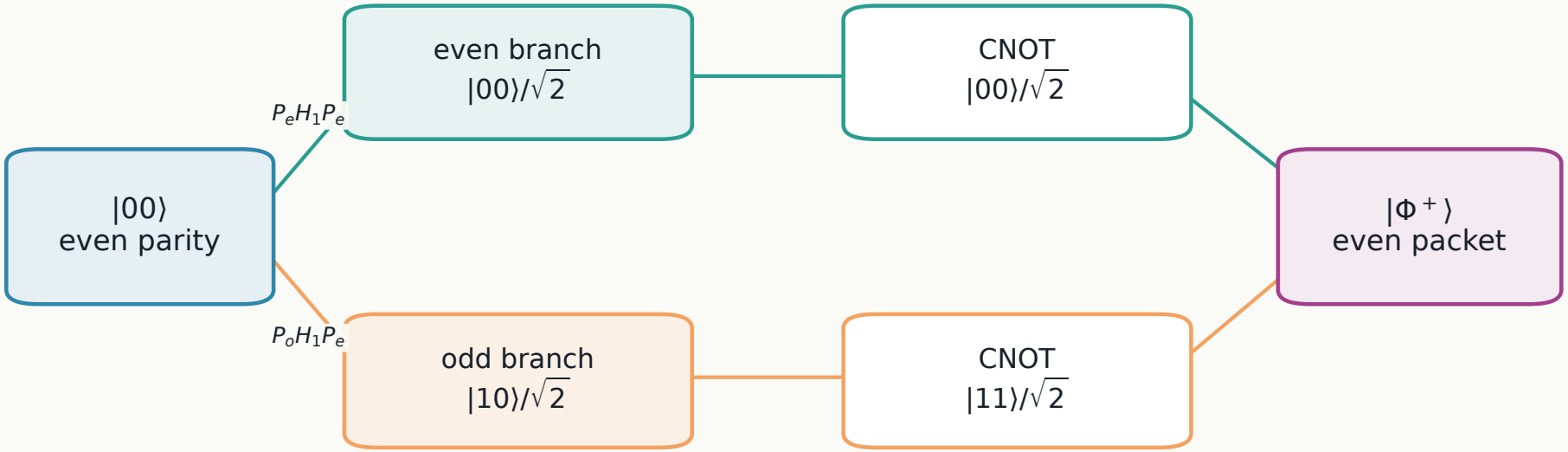
Bell-state preparation as an exact packet path sum



The two nonzero paths synthesize $|00\rangle/\sqrt{2} + |11\rangle/\sqrt{2}$; every other parity path vanishes.